

# Mathematics A

## Unit 1 • Chapter OL-T10 Classified

Cross-session • chapter-organised candidate

13 classified items • 39 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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# Contents

Unit 1 • Unit 1 • Chapter OL-T10 • 13 items • 39 marks

| Chapter          | Items | Marks | Starts |
|------------------|-------|-------|--------|
| 01. OL-T10 Surds | 13    | 39    | 4      |

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

# Surds

Unit 1 • 39 marks • 13 item(s)

June 2025 | 4MA1-1H Q15 | 4 marks

Binomial surd denominator using a conjugate

June 2024 | 4MA1-1H Q17 | 3 marks

Binomial surd denominator using a conjugate

November 2023 | 4MA1-2H Q15 | 3 marks

Single-term surd denominator

June 2023 | 4MA1-2H Q23 | 4 marks

Binomial surd denominator using a conjugate

November 2024 | 4MA1-1H Q17 | 1 marks

Extract square factors from a surd

November 2024 | 4MA1-1H Q17 | 3 marks

Binomial surd denominator using a conjugate

June 2022 | 4MA1-1H Q16 | 3 marks

Add or subtract like surds

January 2022 | 4MA1-2H Q17 | 3 marks

Binomial surd denominator using a conjugate

January 2020 | 4MA1-1H Q17 | 3 marks

Expand binomials containing surds

January 2020 | 4MA1-2H Q20 | 3 marks

Binomial surd denominator using a conjugate

January 2021 | 4MA1-1H Q18 | 3 marks

Expand binomials containing surds

November 2021 | 4MA1-2H Q17 | 3 marks

Binomial surd denominator using a conjugate

May-Nov 2020 | 1H Q19 | 3 marks

Single-term surd denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

**Working Unit 11**

4MA1-1H Q15

ORIGINAL QUESTION

4 marks

15  $(\sqrt{3})^5 = k\sqrt{3}$  where  $k$  is an integer.

(a) Find the value of  $k$

$k = \dots\dots\dots$   
(1)

(b) Show that  $\frac{21}{3 - \sqrt{2}}$  can be written in the form  $c + \sqrt{d}$

where  $c$  and  $d$  are integers.

Show each stage of your working clearly.

(3)

**Worked Solution 11**

Powers and rationalising a surd

**UNIT 1**  
**4 marks****Part (a)**

$$(\sqrt{3})^5 = (\sqrt{3})^4 \sqrt{3} = 9\sqrt{3} \Rightarrow k = 9.$$

**Use the conjugate**

$$\frac{21}{3 - \sqrt{2}} \cdot \frac{3 + \sqrt{2}}{3 + \sqrt{2}} = \frac{21(3 + \sqrt{2})}{9 - 2} = 9 + 3\sqrt{2} = 9 + \sqrt{18}.$$

**Final answer**

$$k = 9, \quad 9 + \sqrt{18}$$

**Question 12**

4MA1-1H Q17

ORIGINAL QUESTION

3 marks

17 Show that  $\frac{1 + \sqrt{5}}{3 - \sqrt{5}}$  can be written in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

Show each stage of your working clearly.

**Worked Solution 12**

Rationalise the denominator

MODULAR UNIT 1

3 marks

**Conjugate**

$$\frac{1 + \sqrt{5}}{3 - \sqrt{5}} \frac{3 + \sqrt{5}}{3 + \sqrt{5}} = \frac{8 + 4\sqrt{5}}{4}.$$

**Final answer**

$$2 + \sqrt{5}$$

**Question 27**

4MA1-2H Q15 (b)

ORIGINAL QUESTION UNIT 1

3 marks

- (b) Express  $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$  in the form  $\sqrt{n}$  where  $n$  is an integer.

Show your working clearly.

.....  
(3)



**Worked solution 27**

Simplify a surd expression

MODULAR UNIT 1

3 marks

**(b) – Simplify a surd expression**

Simplify the numerator surds

$$\sqrt{125} + \sqrt{80} = 5\sqrt{5} + 4\sqrt{5} = 9\sqrt{5}.$$

Rationalise the denominator

$$\frac{9\sqrt{5}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{15}}{3} = 3\sqrt{15}.$$

Put it in the requested form

$$3\sqrt{15} = \sqrt{9 \cdot 15} = \sqrt{135}.$$

**Final answer** $\sqrt{135}$

**Question 27**

4MA1-2H Q23

ORIGINAL QUESTION

4 marks

- 23 The diagram shows a cuboid with a square cross section.

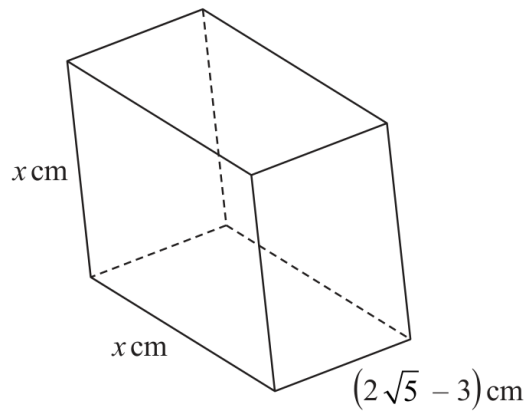


Diagram **NOT**  
accurately drawn

The volume of the cuboid is  $(13 + 6\sqrt{5})\text{ cm}^3$

Without using a calculator, find the value of  $x$

Give your answer in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

Show your working clearly.

$x =$  .....

**Worked Solution 27**

Surd volume and a square cross-section

MODULAR UNIT 1

4 marks

**Form x squared**

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5} \Rightarrow x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}.$$

**Rationalise**

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{11} = 9 + 4\sqrt{5}.$$

**Recognise the square**

$$9 + 4\sqrt{5} = (2 + \sqrt{5})^2.$$

**Take the positive length**

$$x = 2 + \sqrt{5}.$$

**Final answer**

$$2 + \sqrt{5} \text{ cm}$$

**Original question 13**

4MA1-1H Q17 M01

MODULAR UNIT 1

1 marks

17 (a) Express  $\sqrt{675}$  in the form  $n\sqrt{27}$  where  $n$  is a positive integer.

.....  
(1)

**Worked solution 13**

Express a surd in the required form

MODULAR UNIT 1

1 marks

**M01 – Express a surd in the required form****Factor inside the root**

$$\sqrt{675} = \sqrt{25 \times 27} = 5\sqrt{27}.$$

**Final answer**

$$n = 5$$

**Original question 14**

4MA1-1H Q17 M02

MODULAR UNIT 1

3 marks

- (b) Show that  $\frac{5 - \sqrt{2}}{\sqrt{2} - 1}$  can be written in the form  $a + b\sqrt{2}$  where  $a$  and  $b$  are integers.

(3)

**Worked solution 14**

Rationalise a surd denominator

MODULAR UNIT 1

3 marks

**M02 – Rationalise a surd denominator****Multiply by the conjugate**

$$\frac{5 - \sqrt{2}}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}.$$

**Simplify**

The denominator is 1 and the numerator is  $(5 - \sqrt{2})(\sqrt{2} + 1) = 4 + 4\sqrt{2}$ .

**Identify integers**

$$a = 4, b = 4.$$

**Final answer**

$$4 + 4\sqrt{2}$$

**Original question 11**

4MA1-1H Q16 Q16

**ORIGINAL QUESTION****3 marks**

- 16 Without using a calculator, show that  $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$   
Show your working clearly.



**Worked solution 11**

Chapter: Surds

MODULAR UNIT 1

3 marks

**Q16 – Chapter: Surds**

Family: Simplify and collect surds • Variant: Add or subtract like surds

**Method**

$$12/(\sqrt{2} - 1) = 12(\sqrt{2} + 1)/[(\sqrt{2} - 1)(\sqrt{2} + 1)] = 12\sqrt{2} + 12.$$

*Why: Multiply by the conjugate to rationalise the denominator.***Method**

$$(\sqrt{2})^5 = 4\sqrt{2}, \text{ so the left side is } 12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12.$$

*Why: Use powers of  $\sqrt{2}$  and collect like surds.***Method**

Since  $\sqrt{32} = 4\sqrt{2}$ , the right side  $2\sqrt{32} + 12$  is also  $8\sqrt{2} + 12$ ; therefore the identity is shown.

*Why: Simplify both sides to the same exact surd form.***Final answer****shown identity:** Both sides simplify to  $8\sqrt{2} + 12$

**Original question 25**

4MA1-2H Q17

**ORIGINAL QUESTION****3 marks**

17 Show that  $\frac{\sqrt{12}}{\sqrt{3} + 2}$

can be written in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

**Worked solution 25**

Chapter: Surds

MODULAR UNIT 1

3 marks

**Q17 – Chapter: Surds**

Family: Rationalise a surd denominator • Variant: Binomial surd denominator using a conjugate

**Method** $\sqrt{12}/(\sqrt{3}+2) \times (\sqrt{3}-2)/(\sqrt{3}-2)$  rationalises the denominator.*Why: Multiply by the conjugate.***Method**The numerator is  $6-2\sqrt{12}$  and the denominator is  $-1$ .*Why: Expand and simplify the denominator.***Method**The result is  $-6+\sqrt{48}$ .*Why: Divide by  $-1$  and write in the required form.***Final answer****surd form:**  $-6+\sqrt{48}$

17 (a) Show that  $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

## Worked solution

January 2020 | Question 17 | 3 marks

Family: Multiply and expand surd expressions • Variant: Expand binomials containing surds

Method / working

1.  $\sqrt{12} = 2\sqrt{3}$ , so  $6 + 2\sqrt{12} = 6 + 4\sqrt{3}$ .

2. Square:  $(6 + 4\sqrt{3})^2 = 84 + 48\sqrt{3}$ .

3. Factor 12:  $84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$ .

Final answer:  $12(7 + 4\sqrt{3})$

**20** The area of a rectangle is  $18\text{ cm}^2$

The length of the rectangle is  $(\sqrt{7} + 1)\text{ cm}$ .

Without using a calculator and showing each stage of your working,  
find the width of the rectangle.

Give your answer in the form  $a\sqrt{b} + c$  where  $a$ ,  $b$  and  $c$  are integers.

..... cm

## Worked solution

January 2020 | Question 20 | 3 marks

Family: Rationalise a surd denominator • Variant: Binomial surd denominator using a conjugate

Method / working

1.  $\text{Width} = \frac{18}{(\sqrt{7}+1)}$ .
  2. Multiply by the conjugate  $\frac{(\sqrt{7}-1)}{(\sqrt{7}-1)}$ .
  3.  $\frac{18(\sqrt{7}-1)}{(7-1)} = 3\sqrt{7}-3$  cm.
- Final answer:  $3\sqrt{7}-3$  cm

- 18** Given that  $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$  where  $x$  is a prime number and  $y$  is an integer, find the value of  $x$  and the value of  $y$ . Show each stage of your working clearly.

$x =$  .....

$y =$  .....



## Worked solution

January 2021 | Question 18 | 3 marks

Family: Multiply and expand surd expressions • Variant: Expand binomials containing surds

Method / working

1. Expand  $(8-\sqrt{x})(5+\sqrt{x})=40+3\sqrt{x}-x$ .
2. Compare with  $y\sqrt{x}+21$ :  $x-19=0$ , so  $x=19$ .
3. Then  $y=3$ .

Final answer:  $x=19$ ,  $y=3$

17 Express  $\frac{8}{\sqrt{5}-1}$  in the form  $\sqrt{a} + b$  where  $a$  and  $b$  are integers.

Show each stage of your working clearly.

## Worked solution

November 2021 | Question 17 | 3 marks

Family: Rationalise a surd denominator • Variant: Binomial surd denominator using a conjugate

Method / working

1.  $\frac{8}{\sqrt{5}-1}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. Rationalise the denominator:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3.  $\frac{8}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4.  $=\frac{8(\sqrt{5}+1)}{5-1}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5.  $=2(\sqrt{5}+1)=2\sqrt{5}+2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. In the form  $\sqrt{a+b}$ ,

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7.  $2\sqrt{5}+2=\sqrt{20}+2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. Answer:  $\sqrt{20}+2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer:  $\sqrt{20}+2$

**19** Without using a calculator, rationalise the denominator of  $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

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(Total for Question 19 is 3 marks)

## Answer reference | May-Nov 2020 | Question 19

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

|    |                                                                                                                              |               |   |                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------------------|---------------|---|-------------------------------------------------------------------------------------------|
| 19 | $\frac{6}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}}$ or $\frac{6}{3-\sqrt{7}} \times \frac{-3-\sqrt{7}}{-3-\sqrt{7}}$ |               |   | M1                                                                                        |
|    | $\frac{6(3+\sqrt{7})}{3^2-7}$ or $\frac{6(3+\sqrt{7})}{2}$ or $\frac{6(-3-\sqrt{7})}{-3^2+7}$ or $\frac{6(-3-\sqrt{7})}{-2}$ |               |   | M1 (numerator may be expanded or denominator may be 4 terms which need to be all correct) |
|    |                                                                                                                              | $9+3\sqrt{7}$ | 3 | A1 dep on M2 for $9+3\sqrt{7}$ or $3(3+\sqrt{7})$ from correct working                    |
|    |                                                                                                                              |               |   | <b>Total 3 marks</b>                                                                      |

## Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

Method

$$(6)/(3-\sqrt{7})$$

Rationalise the denominator:

$$(6)/(3-\sqrt{7}) \times (3+\sqrt{7})/(3+\sqrt{7})$$

$$=(6(3+\sqrt{7}))/((9-7))$$

$$=(6(3+\sqrt{7}))/2=3(3+\sqrt{7})$$

$$=9+3\sqrt{7}$$

Answer:  $9+3\sqrt{7}$